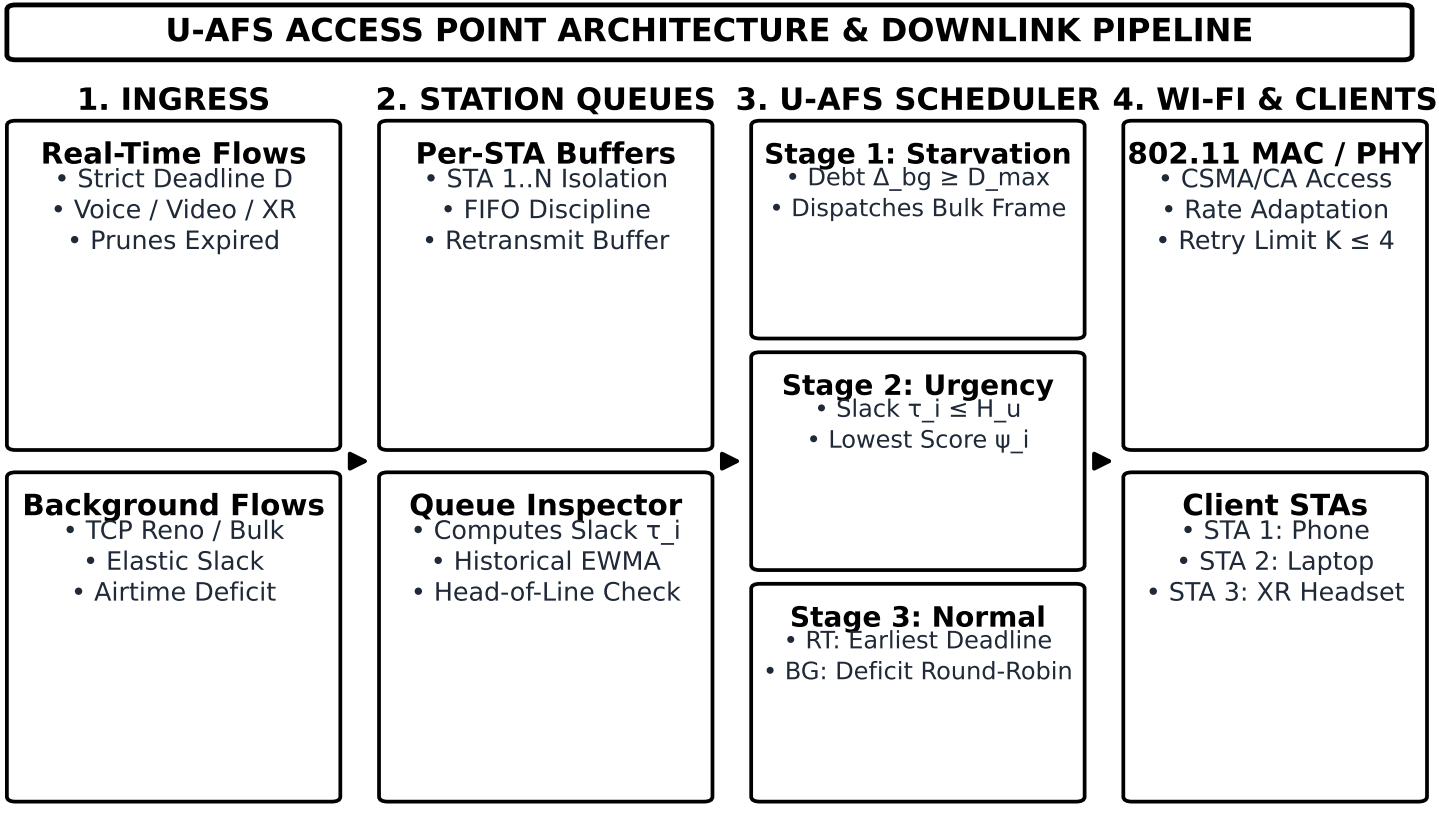
U-AFS ACCESS POINT ARCHITECTURE & DOWNLINK PIPELINE
1. INGRESS
2. STATION QUEUES
3. U-AFS SCHEDULER
4. WI-FI & CLIENTS
Real-Time Flows
• Strict Deadline D
• Voice / Video / XR
• Prunes Expired
Per-STA Buffers
• STA 1..N Isolation
• FIFO Discipline
• Retransmit Buffer
Stage 1: Starvation
• Debt Δ_bg ≥ D_max
• Dispatches Bulk Frame
802.11 MAC / PHY
• CSMA/CA Access
• Rate Adaptation
• Retry Limit K ≤ 4
Stage 2: Urgency
• Slack τ_i ≤ H_u
• Lowest Score ψ_i
Background Flows
• TCP Reno / Bulk
• Elastic Slack
• Airtime Deficit
Queue Inspector
• Computes Slack τ_i
• Historical EWMA
• Head-of-Line Check
Stage 3: Normal
• RT: Earliest Deadline
• BG: Deficit Round-Robin
Client STAs
• STA 1: Phone
• STA 2: Laptop
• STA 3: XR Headset